

Date	30 September 2026	Team ID	SWTID-2026-26
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer	Maximum Marks	4 Marks

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

ARCHITECTURE IMPLEMENTATION GUIDELINES:

- **Application Logic & Processes:** Include all core process blocks (Catalog Engine, Approval Routing, and Flow Designer Engine running the "Standard laptop task" flow under System User).
- **Infrastructural Demarcation:** Explicitly demarcate the presentation tier (Service Portal / Workspace), cloud application tier (Now Platform PaaS/SaaS), and multi-datacenter database storage.
- **External Interfaces:** Identify integration boundaries including ServiceNow REST Table APIs and vendor IntegrationHub Spokes for procurement orders.
- **Data Storage Components:** Specify underlying database tables ('sc_request', 'sc_req_item', 'sysapproval_approver', 'sc_task', and 'sc_cat_item').
- **Machine Learning Models:** Detail ServiceNow AI Search and Predictive Intelligence support for catalog recommendations and search parsing.

TECHNICAL ARCHITECTURE:

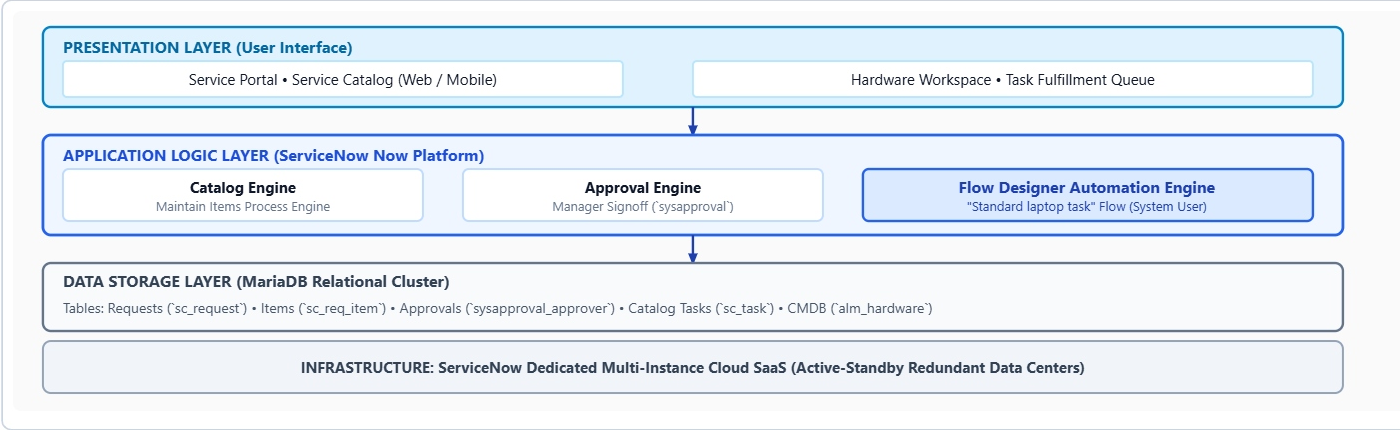


TABLE-1: TECHNOLOGY STACK COMPONENTS:

S.No	Component	Description	Technology
1.	User Interface	Self-service portal interface for laptop selection and tracking; workspace interface for technician queue management.	Service Portal, AngularJS, Bootstrap 3
2.	Application Logic-1	Service Catalog Engine handling order submissions, variable capturing, and generating parent REQ/RITM records.	ServiceNow Catalog Engine, Glide API
3.	Application Logic-2	Automated workflow orchestration executing under System User in Global scope upon Service Catalog submission.	ServiceNow Flow Designer
4.	Application Logic-3	Automated task creation engine mapping Short Description as "Laptop need to Configured" and Assignment Group as "Hardware".	Flow Action: Create Catalog Task
5.	Database	Relational database maintaining integrity across requests, items, approval states, and catalog task assignments.	MariaDB Enterprise Cluster
6.	Cloud Database / Storage	Dedicated multi-instance cloud data storage with isolated enterprise tenancy and automated replication.	ServiceNow Cloud Instance Database
7.	File Storage	Attachment storage for vendor quotes, device staging checklists, and technical documentation.	ServiceNow sys_attachment Table
8.	External API	Inbound/Outbound REST web services for supplier order placement and enterprise asset synchronizations.	ServiceNow REST Table API
9.	Machine Learning / Search	Intelligent portal search helping requesters quickly locate approved standard hardware models.	ServiceNow AI Search Engine
10.	Infrastructure	Cloud SaaS hosting environment with automated failover and secure multi-tier infrastructure.	ServiceNow Multi-Instance Cloud SaaS

TABLE-2: APPLICATION CHARACTERISTICS:

S.No	Characteristics	Description	Technology
1.	Platform & Open Frameworks	Modern front-end rendering combined with robust server-side JavaScript execution for automated processing.	AngularJS, Bootstrap, ECMAScript (Rhino / Node runtime)
2.	Security Implementations	Role-Based Access Control (RBAC) restricting catalog item ordering, manager approvals, and task execution securely under System User.	Access Control Lists (ACLs), TLS 1.3, AES-256 Encryption
3.	Scalable Architecture	Dedicated multi-instance cloud model capable of handling concurrent catalog transactions and flow executions during bulk hiring.	Multi-Instance SaaS, Asynchronous Event Queue
4.	Availability	High-availability active-standby deployment across geographically separated enterprise data center pairs ensuring uninterrupted service.	Advanced HA Architecture, 99.9% Uptime SLA
5.	Performance	Sub-second catalog loading and low-latency automated task generation (under 3 seconds) using optimized query caches.	ServiceNow In-Memory Cache, Flow Background Workers